

MCQs Dr Parsal

Biochemistry (Nucleotides, DNA, RNA and Regulation of Gene Expression)

By Students of 1st Semesters

- 1- The Backbone of DNA is
 - a) Hydrophilic
 - b) Hydrophobic
 - c) Neutral
 - d) Both Hydrophilic and Hydrophobic

- 2- Which is most Stable DNA under normal Physiological condition.
 - a) A-DNA
 - b) B-DNA
 - c) H-DNA
 - d) Z-DNA

- 3- Which of following is true statement?
 - a) DNA is more stable than RNA in Alkaline condition because of the Presence of 3' OH
 - b) DNA is more stable than RNA in Alkaline condition because of the Presence of 2' H compared to RNAs 2' OH
 - c) RNA is more stable than DNA in Alkaline condition because of the Presence of 2' and 3' OH compared to DNA 2' H and 3' OH
 - d) Both are equally Stable in Alkaline conditions

- 4- RNA-DNA Hybrid are
 - a) More Stable than DNA-DNA Hybrid
 - b) More Stable than RNA-RNA Hybrid
 - c) Less Stable than RNA-RNA Hybrid
 - d) Less stable than RNA-RNA Hybrid and DNA-DNA Hybrid

- 5- Type of sugar present in DNA are
 - a) Triose
 - b) Tetrose
 - c) Pentose
 - d) Hexose

- 6- A Short length DNA molecule has 80 thymine and 80 Guanine bases. The Total number of Nucleotide in the DNA fragment is
- a) 160
 - b) 40
 - c) 320
 - d) 640
- 7- Nucleotides in DNA are held by
- A) 3'5' Phosphodiester Bond
 - B) 3'4' Phosphodiester Bond
 - C) 3'2' Phosphodiester Bond
 - D) 3'6' Phosphodiester Bond
- 8- DNA parental Nucleotides sequence is read in direction of 5'-3' and synthesis new strand only in direction
- a) 5'-3' Parallel to reading
 - b) 3'-5' Anti parallel to reading
 - c) Sometime parallel and sometime antiparallel
 - d) It depend upon DNA Polymerase
- 9- Which one is most important to be present for DNA elongation? If it is missing DNA synthesis stop when that nucleotide is depleted.
- a) Deoxyadenosine triphosphate [dATP]
 - b) Deoxythymidine triphosphate [dTTP]
 - c) Deoxyguanosine triphosphate [dGTP]
 - d) Deoxycytidine triphosphate [dCTP]
 - e) All of above are necessary if one is in short supply , stop DNA synthesis
- 10- When DNA is direct to RNA synthesis , During Elongation process Local unwinding of DNA helix is followed by
- a) Synthesis of a 5'-3' RNA transcript coded for by the DNA template read in 3'-5' direction
 - b) Synthesis of a 5'-3' DNA transcript coded for by the RNA template read in 3'-5' direction
 - c) Synthesis of a 3'-5' RNA transcript coded for by the DNA template read in 3'-5' direction
 - d) Synthesis of a 5'-3' RNA transcript coded for by the DNA template read in 5'-3' direction

11- Regulation of Gene expression in Eukaryotes is occur at all below levels but Primarily at

- a) At Transcription
- b) Translation
- c) DNA Level
- d) Co-Post Transcription Level

12- The bond of Nucleoside can be Hydrolyzed by Nucleases but which one of Statement is false

- a) Deoxyribonucleases for DNA
- b) Ribonucleases for RNA
- c) Alkali for DNA only
- d) Alkali for RNA only

13- DNA helicase unwind the DNA strands these two Strands of DNA separated by which of following thing

- a) Protein attached to DNA
- b) By DNA Helicase itself
- c) By DNA Polymerase
- d) DNA Polymerase !!!

14- Nuclear DNA is present in combination with

- a) Histones
- b) Non-histones
- c) Both (A) and (B)
- d) None of these

15- Alkalies cannot hydrolyse

- a) mRNA
- b) tRNA
- c) rRNA
- d) DNA

1- Mode of DNA replication in E.coli is

- a) Conservative and unidirectional
- b) Semiconservative and unidirectional
- c) Conservative and bidirectional
- d) Semiconservative and bidirectional

2. When DNA replication starts

- a) The phosphodiester bonds between the adjacent nucleotides break
- b) The bonds between the nitrogen base and deoxyribose sugar break
- c) The leading strand produces Okazaki fragments
- d) The hydrogen bonds between the nucleotides of two strand break

3- Eukaryotes differ from prokaryote in mechanism of DNA replication due to:

- a) Different enzyme for synthesis of lagging and leading strand
- b) Use of DNA primer rather than RNA primer
- c) Unidirectional rather than bidirectional replication
- d) Discontinuous rather than semidiscontinuous replication

4- _____ joins the DNA fragments to the lagging strand.

A.Telomerase

B.Centromeres

C.Helicases

D.Ligases

E.Antiparallel strands

5. What term is used to describe the process by which DNA is copied to produce two daughter DNA molecules?

- a) reproduction
- b) replication
- c) translation
- d) transcription

6. What term is used to describe the process by which a segment of DNA is copied to produce a molecule of messenger RNA?

- a) reproduction
- b) replication
- c) translation
- d) transcription

7. The adjacent nucleotides are joined by:

a. Ionic bond

b. Hydrogen bond

c. Phosphodiester bond

d. All of these

8. Which of the following statements is true:

a. The two DNA strands are parallel and complementary

b. The two DNA strands are anti parallel and non-complementary

c. The two DNA strands are anti-parallel and complementary

d. None of these

9. DNA is:

a. Positively charged

b. Negatively charged

c. neutral

d. none of these

10. DNA replication takes place in which direction?

A. 3' to 5'

B. 5' to 3'

C. Randomly

D. Vary from organism to organism

1. Each Nucleotide contains:

a. One base

b. two Pentose Sugars

c. one to three Phosphate group and a free amine

d. none of the above

2. The Nucleotides in DNA are linked together by

a. phosphodiester bonds

b. hydrogen bonds

c. ionic bonds

d. hydrophobic interactions

3. The two strands are hold together by
 - a. phosphodiester bonds
 - b. hydrogen bonds
 - c. ionic bonds
 - d. hydrophobic interactions
4. The DNA sequence are read from the
 - a. 3'-end of the chain to the 5'-end (3'→5')
 - b. 3'-end of the chain to the 3'-end (3'→3')
 - c. 5'-end of the chain to the 5'-end (5'→5')
 - d. 5'-end of the chain to the 3'-end (5'→3')
5. Why DNA with more A and T is has lower melting point than the one more in G and C?
 - a. Because there are three hydrogen bonds between G and C but only two between A and T
 - b. Because there are two hydrogen bonds between G and C and three between A and T
 - c. Because there are strong ionic bonds between G and C than A and T.
 - d. Because there Mr of G and C is greater than that of A and T.
6. There are three major structural forms of DNA which are DNA
 - a. A, B, C
 - b. A,B,Z
 - c. Z, Y, A
 - d. A, B, Y
7. They bind to the ssDNA generated by helicases, providing the single-stranded template required by polymerases
 - a. Single-stranded DNA-binding (SSB) proteins
 - b. Dna A proteins
 - c. RNA polymerase
 - d. Ligase
8. Which of the following is a Trans-Acting Elements?
 - a. Promoter
 - b. Operator (Negative role)
 - c. CAP site (Positive role)
 - d. RNA polymerase
9. All of the following are Gene Expression mechanisms in biological cells except
 - a. DNA editing
 - b. Phosphorylation and Methylation
 - c. RNA editing
 - d. Alternative splicing
10. Regulation of Transcription by stringent Response to the amino acid starvation includes
 - a. Lowering the production of ribosomes.
 - b. Increasing RNA polymerase production. Decreasing the usage of amino acids uptake by Alternative splicing

- c. Regulating sugar metabolism.
- d. Expression by Alternative splicing

11- Which of these provide a solution for supercoil of DNA through replication

- A) ligase
- B) topoisomerase
- C) primer
- D) helicase

12- Telomeres are not responsible for

- A) defending DNA against nuclease
- B) making repair systems distinguish the true end of DNA to begin replication
- C) Maintaining the chromosomes' integrity structure
- D) elongating okazaki fragments

1.3 Why is the enzyme DNA ligase required continuously during DNA replication?

- a. The parental strands must be joined back together.
- b. The complex of proteins that work together at the replication fork must be kept from falling apart.
- c. Fragments of the lagging strand must be joined together.
- d. Fragments of the leading strand must be joined together.

14 DNA polymerase extends a polynucleotide strand by doing what?

- a. adding lost DNA sequences to the 3' end.
- b. threading the existing DNA through a replication complex.
- c. building short DNA fragments and linking them together.
- d. covalently linking a single new nucleotide to a previously existing strand.

15. What is the cellular process in which a new copy of DNA is created?

- a. Cell Division
- b Protein Synthesis
- c. Dehydration Synthesis
- d. RNA Synthesis
- e. DNA Replication

16 Why are Okazaki fragments necessary?

- a. Because DNA polymerase can only add new nucleotides to the 3-prime end of a DNA strand.

- b. Because DNA polymerase can not read continuously in either direction.
- c. Because DNA polymerase can only add new nucleotides to the 5-prime end of a DNA strand.
- d. Because DNA polymerase falls off the DNA template very easily.

17- Which enzyme is responsible for binding Okazaki fragments together?

- a. DNA ligase
- b. DNA polymerase
- c. DNA helicase
- d. DNA amylase
- e. RNA primase

18- Both strands of DNA serve as templates concurrently in

- A. replication
- B. excision repair
- C. mismatch repair
- D. none of these

19- Proofreading activity to maintain the fidelity of DNA synthesis

- A. occurs after the synthesis has been completed
- B. is a function of the 3'-5' exonuclease activity of the DNA polymerases
- C. requires the presence of an enzyme separate from the DNA polymerases
- D. occurs in prokaryotes but not eukaryotes

20- . Which of the following repairs nicked DNA by forming a phosphodiester bond between adjacent nucleotides?

- A. Helicase
- B. DNA gyrase
- C. Topoisomerases
- D. DNA ligase

21- The enzyme that catalyzes the synthesis of DNA is called

- A. DNA polymerase
- B. DNA gyrase
- C. DNA ligase
- D. Helicase

22 Which of the following possesses both 5'-3' and 3'-5' exonuclease activity?

- A. Kornberg enzyme
- B. DNA polymerase III
- C. Taq DNA polymerase

D. None of these

23 Which of the following enzymes unwind short stretches of DNA helix immediately ahead of a replication fork?

A. DNA polymerases

B. Helicases

C. Single-stranded binding proteins

D. Topoisomerases

24 The synthesis of DNA by DNA polymerase occurs in the

A. 3' to 5' direction

B. 5' to 5' direction

C. 5' to 3' direction

D. 3' to 3' direction

25 The function of the sigma factor of RNA polymerase is to

A. assure that transcription begins at the proper point

B. assure that transcription ends at the proper point

C. assure that translation begins at the proper point

D. assure that translation ends at the proper point

26 The transcriptional regulatory proteins in eukaryotes which induce transcription usually have

A. at least two distinct domains of protein structure, a DNA binding domain and an activation domain

B. at least two distinct domains of protein structure and a DNA binding domain

C. at least one distinct domains of protein structure, a DNA binding domain and an activation domain

D. none of these

27 In eukaryotes, there are three different RNA polymerases. The RNA polymerase responsible for transcription of mRNA is

A. RNA polymerase I

B. RNA polymerase II

C. RNA polymerase III

D. none of these

28 which of the following is the protein factor identifies the promoter of protein-coding genes in eukaryotes?

A. TATA box

B. Rho

C. sigma

D. TFIID

29- If a codon on the DNA sense strand is ATC, the corresponding tRNA anti-codon might be...

- a) ATC b) AUC c) UAG d) TAG

18. A section of DNA has the following sequence of nitrogenous bases: 5'-CGATTACAG-3'.

Which of the following sequences would be produced as a result of transcription?

- a) 5'-CGTUUTCTG-3'
b) 5'-GCUAAUGUC-3'
c) 3'-GCTAATGTC-5'
d) 3'-GCUAAUGUC-5'

30- At 5'-end of mRNA 7-methylguanosine "cap" attaches through_____

- a. 5'-5' triphosphate linkage
b. 5'-3'triphosphate linkage
c. 5'-3'diphosphate linkage
d. 5'-5'diphosphate linkage

31- In prokaryotes RNA polymerase synthesizes all of the RNA except

- A. tRNA
B. mRNA
C. primer
D. dimer

32- DNA gyrase is

- a. DNA helicase
b. DNA Topoisomerase Type I
c. DNA Topoisomerase Type II
d. DNA Topoisomerase Type III
e. DNA Polymerase

33- Total amount of purines equals the total amount of:

- a. Pyrimidine
b. Thymine
c. Cytosine
d. Adenine

34- The enzymes involved in the DNA replication process are called:

a. semiconservative replication

b. template-directed

c. origin of replication

d. positive supercoils

35. In prokaryotic organisms, DNA replication begins at a single, unique nucleotide sequence—a site called:

a. origin of replication

b. replication bubble

c. bidirectional region

d. replication fork

35. Ribozymes are...

a. catalytic RNA

b. catalyze ribosome synthesis

c. remove exons hnRNA

d. responsible enzyme in ribosome degradation

36. Which enzyme has proofreading activity?

a. RNA polymerase

b. DNA polymerase

c. Helicase

d. Topoisomerase

37- Which enzyme are responsible for removing supercoils in the helix?

a. DNA topoisomerases

b. RNA polymerase

c. DNA polymerase

d. replication fork

38 Phosphodiester bonds of DNA are cleaved by

a) Ribonucleases

b) Topoisomerases

c) Deoxyribonucleases

d) Polymerases

39 The total amount of purines equals the total amount of pyrimidines is a statement of

a) Watson and Crick

a) Chargaff

b) Bohr

c) None

40 Temperature at which half of helical structural is lost is known as

a) Curie's Temperature

b) Critical Temperature

c) Melting Temperature

d) Both 'a' and 'c'

41 Structural form of DNA described by Watson and Crick was

a) A-Form

b) Z-Form

c) B-Form

d) All the above

42 Choose the correct statement

a) Z-DNA is a right handed helix with 12 basepairs per turn

b) Z-DNA is a left handed helix with 12 basepairs per turn

c) Z-DNA is a right handed helix with 10 basepairs per turn

d) Z-DNA is a left handed helix with 10 basepairs per turn

43 When handedness of double and superhelix are same, type of topoisomerism is

a) Relaxed

b) Negative

c) Positive

d) Supertwist

44 The most tissue specific and species specific histone protein is

a) H1

b) H2A

c) H3 & H4

d) H2B

45 Unique target of Fluoroquinolones like ciprofloxacin is

a) Human type 1 topoisomerase

b) Human type 2 topoisomerase

c) DNA B

d) Bacterial DNA gyrase

46- 'RNAi' stands for which of the following

a) RNA inducer

b) RNA insertion

c) RNA interference

d) RNA intron

47 Suppose a certain gene contains the double stranded sequence

5'- ATGTTTAGCGCC-3'

3'- TACAAATCGCGG-5'

If the top strand is the sense strand and codes for an mRNA whose sequence begins 'ATG' which of the following would be the sequence of the corresponding segment of antisense RNA?

a) 5'-AUGUUUAGCGCC-3'

b) 5'-CCGCGAUUUGUA-3'

c) 5'-GGCGCUAAACAU-3'

d) 5'-UACAAAUCGCGG-3'

48 Patients with Xeroderma Pigmentosum cannot perform which of the following repair

a) Base excision repair

b) Mismatch repair

c) Nucleotide excision repair

d) Repair of double strand breaks

49 The process of removing introns and joining exons is called

a) Capping

b) Splicing

c) Tailing

d) Methylation

50 Proof reading or editing activity that can remove an incorrect aminoacid from the tRNA molecule is done by

a) Aminoacyl tRNA synthetase

b) RNA Polymerase 1

c) DNA ligase

d) Primase

51 The codon 5'-AUG-3' designates

a) Valine

b) Alanine

c) Threonine

d) Methionine

51 Alpha amanitin produced by the 'death cap' forms a tight complex with

a) RNA pol II

b) RNA pol III

c) DNA pol III

d) DNA pol II

52 Polyadenylate polymerase is responsible for

a) removal of chains of Adenine

b) response elements that bind STFs

c) removal of mismatched adenine

d) transcription of poly-A tail

53 TATA Box is found in

a) RNA pol III

b) RNA pol II

c) DNA pol III

d) DNA pol II

54 The ZYA region of the Lac operon will be maximally expressed if

- a) cyclic AMP levels are low
- b) glucose and lactose are both available
- c) the attenuation stem loop is able to form
- d) the CAP site is occupied

55 The mechanism by which tRNAs can recognize more than one codon for a specific amino acid is described by

- a) Drifty gene hypothesis
- b) Wobble hypothesis
- c) Degeneracy
- d) Frame shift mutation

56 E site on the ribosome during translation binds to

- a) RNA polymerase
- b) Peptidyl tRNA
- c) Empty tRNA
- d) Aminoacyl tRNA

57 Operon is 'ON' when

- a) + Glucose, - Lactose
- b) - Glucose, + Lactose
- c) + Glucose, + Lactose
- d) - Glucose, - Lactose

58 Iron transporting is

- a) Albumin
- b) Transferrin
- c) Globulin
- d) Transferrin

59 Decrease in the positive charge of histone protein is done by

- a) Histone acetyl transferase

b) Histone deacetyl transferase

c) Methyl transferase

d) Dihydrofolate reductase

60- Kinases are activated in response to the following environmental conditions EXCEPT

a) aminoacid starvation

b) heme deficiency in erythroblasts

c) accumulation of misfolded proteins in RER

d) presence of single stranded DNA

61- Which of the following is best described as cis acting?

a) Cyclic AMP

b) Operator

c) Repressor protein

d) Thyroid hormone nuclear receptor

A purine nucleotide is

(A) AMP (B) UMP

(C) CMP (D) TMP

62- A pyrimidine nucleotide is

(A) GMP

(B) AMP

(C) CMP

(D) IMP

63. Adenine is

(A) 6-Amino purine

(B) 2-Amino-6-oxypurine

(C) 2-Oxy-4-aminopyrimidine

(D) 2, 4-Dioxypyrimidine

64. 2, 4-Dioxypyrimidine is

- (A) Thymine
- (B) Cytosine

(C) Uracil

- (D) Guanine

65. The chemical name of guanine is

- (A) 2,4-Dioxy-5-methylpyrimidine

(B) 2-Amino-6-oxypurine

- (C) 2-Oxy-4-aminopyrimidine

- (D) 2, 4-Dioxypyrimidine

66. Nucleotides and nucleic acids concentration are often also expressed in terms of

- (A) ng

- (B) mg

- (C) meq

(D) OD at 260 nm

67. The pyrimidine nucleotide acting as the high energy intermediate is

- (A) ATP

- (B) UTP

(C) UDPG

- (D) CMP

68. The carbon of the pentose in ester linkage with the phosphate in a nucleotide structure is

- (A) C1

- (B) C3

- (C) C4

(D) C5

69. Uracil and ribose form

(A) Uridine

- (B) Cytidine

(C) Guanosine

(D) Adenosine

70. The most abundant free nucleotide mammalian cells is

(A) ATP

(B) NAD

(C) GTP

(D) FAD